

ReSLLM: Large Language Models are Strong Resource Selectors for Federated Search

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Abstract

We investigate the use of Large Language Models (LLMs) for resource selection in federated search, introducing ReSLLM, a novel prompting approach.

Our ReSLLM method exploits LLMs to drive the selection of resources in federated search. The ReSLLM prompt uses solely information about each resource (e.g. their name) to decide on the suitability of a resource. ReSLLM can be applied in a zero-shot setting. In addition we devise an unsupervised fine tuning protocol, the Synthetic Label Augmentation Tuning (SLAT), where the relevance of previously logged queries and snippets from resources is predicted using an off-the-shelf LLM and then in turn used to fine-tune ReSLLM with respect to the task of resource selection.

Our empirical evaluation and analysis details the factors influencing the effectiveness of LLMs in this context. The results showcase the merits of ReSLLM for resource selection: not only competitive effectiveness is achieved in the zero-shot setting, but also large gains are obtained when relying on the SLAT protocol for fine-tuning. Full results and code for reproduction are available at <https://github.com/ielab/SLAT-RsLLM>.

CCS Concepts

• **Information systems** → **Search engine architectures and scalability**.

Keywords

Large Language Models, Ranking, Prompt

ACM Reference Format:

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1 Introduction

Federated search integrates results from multiple search engines or databases, providing a unified view of dispersed information [20]. This technique enables users to execute a single query across diverse sources, receiving aggregated results that capture a broader spectrum of relevant data. A critical component is *resource selection*, determining which search engines to query based on user intent. Effective resource selection enhances relevance, reduces API costs, and limits queries to the most pertinent resources [16].

We forecast that resource selection will be vital for LLM-based Retrieval-Augmented Generation (RAG) agents, such as langchain¹, llamaindex [13], and DSPy [11]. These agents incorporate retrieval modules, reducing hallucinations and improving reliability, yet face high latency when aggregating results via LLMs, necessitating effective resource selection. Traditional methods rely on human-annotated datasets [7].

This paper introduces ReSLLM, a prompting method for LLMs that selects resources using only search engine descriptions, requiring no training (*zero-shot*). We also present Synthetic Label Augmentation Tuning (SLAT), a fine-tuning protocol using logged queries and synthetic relevance labels generated by LLMs, without the need of human annotated labels.

Experiments on TREC FedWeb 13 and 14 demonstrate that zero-shot ReSLLM surpasses unsupervised baselines, while SLAT-tuned ReSLLM achieves performance comparable to supervised methods. Notably, while the backbone model size and architecture affect zero-shot performance, their influence diminishes with SLAT-tuning.

2 Related Work

Federated search faces challenges in selecting relevant resources, solutions to this are categorized as unsupervised or supervised.

Unsupervised Methods: Methods avoid human labels and rely on lexicon-based [1] or sample-based approaches like ReDDE and SUSHI [21, 24], using Centralized Sample Indexes. Semantic-based methods [8] leverage embeddings but often miss textual nuances.

Supervised Methods: Methods including SVMrank and LTRRS [3, 28] require relevance labels to train models. Graph neural networks [7] improve query-resource modeling but need training data.

Other Methods: Shard selection [15] offers insights applicable to federated search.

¹<https://www.langchain.com/>

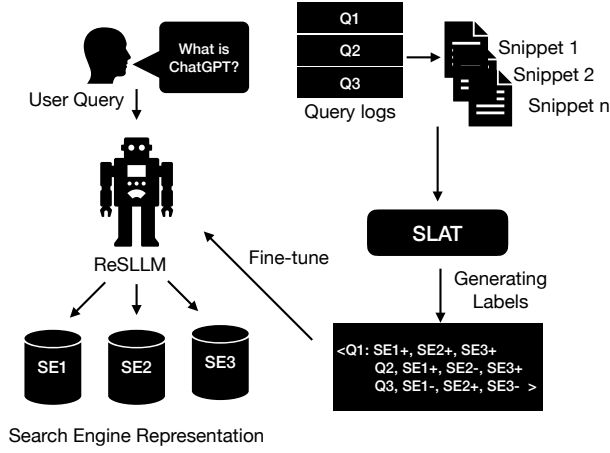


Figure 1: ReSLLM architecture, including the SLAT tuning protocol.

LLMs excel in zero-shot ranking using Pointwise, Listwise, Pairwise, and Setwise approaches [17, 18, 22, 27, 31, 32]. We adopt Pointwise approach, using LLM to predict resource relevance as "yes" or "no." LLM-generated relevance labels [25] align closely with human judgments, with some studies showing GPT-4-generated labels surpassing human annotations. These methods inform our document relevance assessments.

3 Methodology

ReSLLM leverages LLMs as text-based resource selectors to identify resources for query routing in federated search. Figure 1 provides an overview of ReSLLM, including resource representation and fine-tuning via the SLAT protocol.

3.1 Problem Formulation

The resource selection problem for federated search is defined as follows. Given a set R containing n resources ($R = \{r_1, \dots, r_n\}$) and a user query q , an effective resource selector S would select a suitable subset of resources $S(q)$ that best answer q . In practice, search engine practitioners implementing a federated search engine might have constraints on the maximum number of selected resources. We reformulate the resource selection problem so that $S(q)$ needs to be ranked in decreasing likelihood of relevance. We consider scenarios where the involved resources are *uncooperative*, providing no statistical properties but allowing access to a log $\mathcal{L}$ of previously submitted queries and their retrieved snippets.

3.2 The ReSLLM Prompting Method

ReSLLM uses instruction-tuned LLMs for resource selection without task-specific training. Prompts consist of [System Instruction, Resource Representation, User Input, Task Instruction] are demonstrated at Table 1. Resource representations include names, URLs, or descriptions, while user input comprises queries and optionally similar snippets. The

ReSLLM evaluates resources using logits for "yes" and "no" responses. The score is computed as:

Table 1: The ReSLLM prompt. In curly brackets are resource and input variables, e.g., the name of the search engine and the user's query. In the experiments we study the effect of including (or not) the description/snippets. The parts of the prompt that are added (removed) when (not) considering description/snippets are marked in underlined text.

	Prompt
System Instruction	Federated search retrieves information from a variety of sources via a search application built on top of one or more search engines. A user makes a single query request. The federated search then selects only the search engines that the query should be sent to from a list of search engines, and aggregates the result for presentation of high quality result to the user. The task is called resource selection.
Resource Representation	The following is a search engine with its name, url and description. Name: { <i>name</i> } URL: { <i>url</i> } Description: { <i>description</i> }
User Input	The following is a real user query: Query: { <i>query</i> } The following are some snippets from this search engine that are similar to the user query: Snippets: { <i>similar_sampled_snippets</i> }
Task Instruction	Now, please reply only yes or not to indicate if the query should be sent to the search engine. Response:

$$score(q, r_i) = P(\text{yes}|q, r_i) - P(\text{no}|q, r_i) \quad (1)$$

Resources are ranked by $score(q, r_i)$, with optional exclusion of resources with negative scores.

3.3 The SLAT Protocol

SLAT fine-tunes ReSLLM with synthetic labels from query logs, removing the need for human annotations. It includes four steps:

1. *Automatic Relevance Assessment*: SLAT uses LLMs to generate synthetic relevance labels for query-snippet pairs from $\mathcal{L}$. Following Thomas et al. [25], we adapt prompts for open-source LLMs with a five-level relevance scale (Navigational, Key, Highly Relevant, Relevant, Non-Relevant) to align with TREC FedWeb tracks, we show prompt used at Table 2.

2. *Aggregation of Relevance to the Resource Level*: Resource relevance is estimated using graded precision over the top 10 query-snippet pairs [4], with TREC FedWeb 2013 weights: $w_{\text{Non}} = 0$, $w_{\text{Rel}} = 0.25$, $w_{\text{HRel}} = 0.5$, $w_{\text{Key}} = w_{\text{Nav}} = 1$. Aggregated scores are scaled to 0–100.

3. *Creation of Synthetic Training Labels*: Resources are categorized into three relevance levels:

- *Highly Relevant*: Aggregated scores ≥ 50 , labeled "yes" twice.
- *Marginally Relevant*: Scores 25–49, labeled "yes" and "no."
- *Not Relevant*: Scores < 25 , labeled "no" twice.

4. *ReSLLM Fine-tuning*: Synthetic labels are used to fine-tune ReSLLM with contrastive loss. Training queries populate the prompt to fine-tune LLMs for resource selection tasks².

Table 2: Prompt used for instructing the LLM to generate relevance assessments; prompt adapted from [25].

System Prompt	Prompt
	Given a query and a web snippet, you must provide a score on an integer scale of 0 to 4 with the following meanings: 4 = navigational, this page represents a home page of an entity directly named by the query; the user may be searching for this specific page or site. 3 = top relevance, this page or site is dedicated to the topic; authoritative and comprehensive, it is worthy of being a top result in a web search engine. 2 = highly relevant, the content of this page provides substantial information on the topic. 2 = highly relevant, the content of this page provides substantial information on the topic. 1 = minimal relevance, the content of this page provides some information on the topic, which may be minimal; the relevant information must be on that page, not just promising-looking anchor text pointing to a possibly useful page. 0 = not relevant, the content of this page does not provide useful information on the topic, but may provide useful information on other topics, including other interpretations of the same query.
User Prompt	Assume that you are writing a report on the subject of the topic. mark the web snippet according to the previous scale description. Query: A person has typed {query} into a search engine. Result: Consider the following web snippet: —BEGIN WEB Snippet CONTENT— {snippet} —END WEB Snippet CONTENT— Instructions: Split this problem into steps: Consider the underlying intent of the search. Measure how well the content matches a likely intent of the query (M) Measure how trustworthy the web page is (T). Consider the aspects above and the relative importance of each, and decide on a final score (O). Produce a JSON of scores without providing any reasoning. Example:{"M": 2, "T": 1,"O": 1} Results:

4 Experimental Setup

4.1 Considered LLMs

We evaluate seven LLMs of varying architectures and sizes:

- *Flan-T5 Models* [2]: Instruction-tuned T5 models (flan-t5-large: 783M, flan-t5-xl: 2.8B, flan-t5-xxl: 11.3B). Flan-t5-xl is the default.
- *LlaMa2 Models* [26]: Chat-tuned decoder-only models (LlaMa2-7b-chat: 7B, LlaMa2-13b-chat: 13B).
- *Mixtral_7Bx2_Moe*²: A 12.9B mixture of experts model.
- *SOLAR-10.7B-Instruct-v1.0*³: Scaled to 10.7B parameters using depth up-scaling [12].

4.2 Datasets and Evaluation Measures

We use TREC Federated Web collections *FedWeb-13* and *FedWeb-14*: *FedWeb-13* has data from 157 search engines, 50 test queries, and 2,000 synthetic queries with snippets [4]; *FedWeb-14* has data from 149 search engines and 4,000 synthetic queries [5].

The *Greatest Hit* adds 406 queries for SLAT training and embedding baseline creation. Ad-hoc queries are used directly, while conversational queries are derived from FedWeb descriptions.

Evaluation uses nDCG@10, 20, 100, and nP@1, 5, following TREC FedWeb conventions. nP@k evaluates graded relevance at cut-off k , normalized by the ideal achievable score.

²https://huggingface.co/cloudyu/Mixtral_7Bx2_MoE

³<https://huggingface.co/upstage/SOLAR-10.7B-Instruct-v1.0>

4.3 Baselines

Supervised Baseline: *ECNUCS-ecomsvz* [10], the top nDCG@20 run from FedWeb 2014, leveraging human labels and external APIs.

Unsupervised Baseline: *UPD-UPDFW13mu* [6] (TWF-IRF weighting) and *Dragon-drexelRS7* [30] (SUSHI method using query logs).

Embedding Baseline: BAAI/bge-large-en-v1.5 [29] encodes queries and snippets, scoring with cosine similarity.

4.4 SLAT Settings

Synthetic Labels: SOLAR-10.7B-Instruct-v1.0 generates relevance labels. Agreement with TREC assessors yields $\kappa = 0.35$, compared to $\kappa = 0.6$ between assessors.

Conversational Query Generation: Conversational queries are generated from logged queries and snippets using SOLAR-10.7B-Instruct-v1.0⁴.

Fine-tuning Settings: Flan-large, Flan-xl, and LLaMa2-7b-chat are fine-tuned using PyTorch Lightning and Alpaca guidelines [23]. Training spans 1–2 epochs with contrastive loss, optimizing batch sizes for GPU memory.

5 Results and Analysis

5.1 Zero-shot ReSLLM

We evaluate zero-shot ReSLLM using resource representations based on name and URL. Table 3 compares its performance to baselines.

Key Findings:

- ReSLLM’s effectiveness depends heavily on the LLM backbone. Solar-11b outperforms other LLMs across most metrics, though Mixtral-13b achieves marginally better results in certain cases.
- Solar-11b and Mixtral-13b significantly outperform the Embedding baseline, while other LLMs (e.g., Flan-xl, LLaMa2-7b-chat) are less effective.
- ReSLLM outperforms unsupervised baselines but remains less effective than the supervised baseline, *ECNUCS-ecomsvz*, which relies on annotated labels and external API support.
- Conversational queries yield better results than ad-hoc queries, likely due to richer user intent information.

In summary, ReSLLM surpasses unsupervised and embedding baselines, though supervised baselines remain superior.

5.2 ReSLLM with SLAT

Fine-tuning ReSLLM using the SLAT protocol shows notable performance improvements, as shown in Table 4.

Key Findings:

- SLAT+ReSLLM achieves the highest effectiveness across most metrics, approaching or surpassing the supervised baseline in FedWeb-14.
- A surprising gap in nP@1 for FedWeb-14 suggests potential limitations in SLAT’s pseudo-labeling thresholds, affecting top-ranked resource differentiation.
- Conversational queries show no significant improvement over ad-hoc queries post-SLAT fine-tuning, possibly due to mismatches between generated labels and conversational intent.

⁴Table 3 at <https://github.com/ielab/SLAT-RsLLM> for prompt.

Table 3: Effectiveness of zero-shot ReSLLM method in comparison with baseline approaches. * indicates statistical significant differences against the Embedding Baseline (two-tail paired t-test, $p < 0.05$). Effectiveness for TREC baselines for conversational queries is not attainable, as the track did not consider these queries.

	Method	LLM	Ad-hoc					Conversational				
			ndcg10	ndcg20	ndcg100	nP1	nP5	ndcg10	ndcg20	ndcg100	nP1	nP5
FedWeb-13	Unsupervised Baseline		-	0.2990	-	0.1600	0.2100	-	-	-	-	-
	Embedding Baseline		0.2067	0.2508	0.5088	0.1611	0.1880	0.2789	0.3398	0.5608	0.2259	0.2481
	ReSLLM	flan-xl	0.1511*	0.2054*	0.4123*	0.1239	0.1374*	0.1789*	0.2143*	0.4384*	0.1939	0.1638*
	ReSLLM	llama2-7b-chat	0.0683*	0.0815*	0.2626*	0.0273*	0.0644*	0.1156*	0.1467*	0.3599*	0.0780*	0.1045*
	ReSLLM	solar-11b	0.2636*	0.3029	0.5351	0.2713*	0.2486	0.3573*	0.3944	0.5982	0.3809*	0.3398
	ReSLLM	mixtral-13b	0.2605	0.3059	0.5224	0.1897	0.2492	0.2802	0.3246	0.5562	0.2392	0.2726
	ReSLLM	llama2-13b-chat	0.1015*	0.1339*	0.3036*	0.1045	0.1069*	0.1640*	0.2140*	0.4129*	0.1733	0.1564*
FedWeb-14	Unsupervised Baseline		0.3590	0.4220	-	0.2930	0.3140	-	-	-	-	-
	Supervised Baseline		0.6240	0.7120	-	0.5350	0.6040	-	-	-	-	-
	Embedding Baseline		0.1928	0.2418	0.5041	0.1940	0.1740	0.2680	0.3364	0.5568	0.2271	0.2489
	ReSLLM	flan-xl	0.1484*	0.1845*	0.3799*	0.1097	0.1377	0.1801*	0.2092*	0.4087*	0.1717	0.1866
	ReSLLM	llama2-7b-chat	0.0529*	0.0717*	0.2500*	0.0232*	0.0441*	0.0606*	0.0976*	0.3132*	0.0355*	0.0594*
	ReSLLM	solar-11b	0.4093*	0.4227*	0.6115*	0.3931*	0.3926*	0.4351*	0.4509*	0.6362*	0.3803*	0.4363*
	ReSLLM	mixtral-13b	0.2870	0.3431	0.5614	0.2896	0.2572	0.2927	0.3558	0.5680	0.2980	0.2653
	ReSLLM	llama2-13b-chat	0.0977*	0.1184*	0.3370*	0.1047	0.0856*	0.1599*	0.1945*	0.4008*	0.1987	0.1401*

Table 4: Effectiveness of SLAT+ReSLLM, in comparison to the zero-shot ReSLLM and the baselines. * indicates statistical significant differences against the Embedding Baseline (two-tail paired t-test, $p < 0.05$). Effectiveness for TREC baselines for conversational queries is not attainable, as the track did not consider these queries.

	Method	LLM	Ad-hoc					Conversational				
			ndcg10	ndcg20	ndcg100	nP1	nP5	ndcg10	ndcg20	ndcg100	nP1	nP5
FedWeb-13	Unsupervised Baseline		-	0.2990	-	0.1600	0.2100	-	-	-	-	-
	Embedding Baseline		0.2067	0.2508	0.5088	0.1611	0.1880	0.2789	0.3398	0.5608	0.2259	0.2481
	ReSLLM	flan-xl	0.1511	0.2054	0.4123	0.1239	0.1374	0.1789	0.2143	0.4384	0.1939	0.1638
FedWeb-14	SLAT+ReSLLM	flan-xl	0.7557*	0.7897*	0.8574*	0.6929*	0.7752*	0.7556*	0.7988*	0.8622*	0.6950*	0.7857*
	Unsupervised Baseline		0.3590	0.4220	-	0.2930	0.3140	-	-	-	-	-
	Supervised Baseline		0.6240	0.7120	-	0.5350	0.6040	-	-	-	-	-
FedWeb-14	Embedding Baseline		0.1928	0.2418	0.5041	0.1940	0.1740	0.2680	0.3364	0.5568	0.2271	0.2489
	ReSLLM	flan-xl	0.1484	0.1845	0.3799	0.1097	0.1377	0.1801	0.2092	0.4087	0.1717	0.1866
	SLAT+ReSLLM	flan-xl	0.6582*	0.7088*	0.7855*	0.3060*	0.6854*	0.6557*	0.7052*	0.7833*	0.3060*	0.7008*

SLAT+ReSLLM highlights the potential of LLM-generated labels for fine-tuning, rivaling supervised baselines without human-annotated data, whereas current supervised methods often rely on extensive-human annotations.

5.3 Impact of LLM Size and Architecture

The impact of LLM size and architecture on ReSLLM’s performance is shown in Figure 1 and Figure 2 at Github Appendix, respectively. Figure 1 illustrates how increasing the size of the LLM improves effectiveness in both zero-shot and SLAT fine-tuned settings. Figure 2 highlights differences in performance between encoder-decoder and decoder-only models, both before and after fine-tuning.

Model Size: Increasing LLM size enhances zero-shot effectiveness, with nDCG@20 improving by 32% (Flan-large to Flan-xl) and 60% (Flan-xl to Flan-xxl). Post-SLAT fine-tuning, the performance

gap diminishes, indicating size is critical in zero-shot but less so after fine-tuning.

Model Architecture: Encoder-decoder models (e.g., Flan-xl) outperform decoder-only models (e.g., LLaMa2-7b-chat) in zero-shot settings. Post-SLAT fine-tuning, architectural differences are mitigated, suggesting fine-tuning aligns models for downstream tasks.

5.4 Impact of Resource Representation

The effect of different resource representations on ReSLLM’s performance is summarized in Figure 3, available at Github Appendix. We evaluate name+URL (N), description (D), and similar snippets (S) representations, both individually and in combination.

Key Findings:

- In zero-shot, richer representations (N+D, N+S) modestly improve effectiveness (+6%, +3.5%), while combining all (N+D+S) offers marginal gains (+1.1%).
- Post-SLAT fine-tuning, adding similar snippets (N+S) reduces effectiveness (−35%), highlighting its limited utility in this setting.

Overall, careful selection of resource representations is critical, especially post-fine-tuning, to optimize ReSLLM’s effectiveness.

6 Conclusion and Discussion

This study introduces ReSLLM, a zero-shot LLM-based resource selection method for federated search, and SLAT, a fine-tuning protocol leveraging LLM-generated synthetic labels. ReSLLM and SLAT+ReSLLM outperform unsupervised and embedding baselines and, in some cases, rival supervised models that rely on human annotations.

Our findings highlight the potential of ReSLLM for improving RAG pipelines, particularly in conversational agents. Effective resource selection is crucial for enhancing the quality of downstream tasks. Although this work does not directly evaluate RAG applications, it lays a foundation for future studies, including the creation of federated search datasets tailored to RAG pipeline evaluation.

LLMs demonstrate strong performance across NLP tasks, but concerns remain about the inclusion of benchmark data in their training corpora, particularly for undisclosed sources [9, 14]. In our study, ReSLLM’s zero-shot performance was lower than fine-tuned versions, indicating that training data containment does not inherently guarantee higher effectiveness. Fine-tuning proved essential for optimal results.

Future work should explore the impact of prompt variability on resource selection effectiveness. While prompt variability affects zero-shot performance [19], its influence diminishes with fine-tuning. Investigating this aspect could further refine resource selection methodologies and broaden the applications of ReSLLM in federated search and RAG pipelines.

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